

C4MAT1T1 – E – 26 – 7240



FOURTH SEMESTER B.SC. (REVISED NEP) DEGREE EXAMINATION, JUNE 2026

MATHEMATICS (DSC – 1)

Differential Equations and Integral Transforms

Time : 3 Hours]

[Max. Marks : 80

I. Answer **all** questions.

(10×2=20)

- 1) What is exact differential equation ?
- 2) Solve $P^2 - 7P + 12 = 0$.
- 3) Solve $y = px + p - p^2$.
- 4) Form the partial differential equation by eliminating arbitrary function f from $z = f(x^2 + y^2)$.
- 5) Find the complete solution of $\sqrt{p} + \sqrt{q} = 1$.
- 6) Solve $2r + 5s + 2t = 0$.
- 7) Find the Laplace transform of $\sin^2 2t$.
- 8) Define unit step function and write its Laplace transform.
- 9) Define Fourier series of periodic function $f(x)$ with period 2π .
- 10) Find the Fourier coefficient a_0 for the function $f(x) = e^{-ax}$ in $(-\pi, \pi)$.

II. Answer **any six** of the following.

(6×5=30)

- 11) Solve $x \frac{dy}{dx} + y = y^2 \log x$.
- 12) Solve $(D^2 + 2D + 1)y = x \sin x$.
- 13) Derive the partial differential equation $Pp + Qq = R$ by eliminating arbitrary function ϕ from $\phi(u, v) = 0$, where u and v are the functions of x, y and z .
- 14) Solve $(p^2 + q^2)y = qz$ by Charpit's method.
- 15) Find the inverse Laplace transform of
 - i) $\frac{S+12}{S^2+4S}$
 - ii) $\frac{2S+3}{(S-1)(S+2)}$
- 16) If $L[f(t)] = F(S)$, then prove that $L[f''(t)] = S^2F(S) - Sf(0) - f'(0)$.

[P.T.O.]



17). Obtain the Fourier series of $f(x) = x^2$ in $(-\pi, \pi)$ and show that

$$\frac{\pi^2}{4} = 1 + \frac{1}{4} + \frac{1}{9} + \dots$$

18). Find Fourier finite cosine transform of $f(x) = 2 - x$ in $(0, 2)$.

III. Answer **any three** of the following.

(3×10=30)

19) a) With usual notation prove that $\frac{1}{f(D)}(e^{ax}V) = e^{ax} \frac{1}{f(D+a)}V$, where V is a function of x .

b) Solve $(x^2y^2 + xy + 1)ydx + (x^2y^2 - xy + 1)x dy = 0$.

20) a) Explain Charpit's method for solving the partial differential equation $f(x, y, z, p, q) = 0$.

b) Solve $(D^2 - 5DD' + 4D'^2)z = \sin(4x - y)$.

21) a) Prove that the Laplace transform of a periodic function $f(t)$ with period T is

$$L[f(t)] = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

b) Evaluate $L\left[\frac{\cos at - \cos bt}{t}\right]$.

22) a) Find the Fourier series for the function $f(x) = \begin{cases} 1-x & \text{if } -\pi < x \leq 0 \\ 1+x & \text{if } 0 < x \leq \pi \end{cases}$ of period 2π .

b) Solve using Laplace transform $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^{3t}$ given $y(0) = 0 = y'(0)$.

FOURTH SEMESTER B.Sc. (R-NEP-DSC) DEGREE EXAMINATION, AUGUST 2026

Mathematics

DIFFERENTIAL EQUATIONS AND INTEGRAL TRANSFORMS

Time : Three Hours

Maximum : 80 Marks

*Answer all the questions.*I. Answer *all* questions :

- 1 Solve : $p^2 - 5p + 6 = 0$.
- 2 Verify whether the differential equation :
 $(\cos y + y \cos x) dx + (\sin x - x \sin y) dy = 0$ is exact.
- 3 Form the partial differential equation by eliminating arbitrary constants a and b from
 $z = (x+a)(y+b)$.
- 4 Find the complete integral of $p^2 - q^2 = 1$.
- 5 Find the Laplace transformation of $e^{4t} t^5$.
- 6 Define Dirac delta function and write its Laplace transformation.
- 7 Find the inverse Laplace transformation of $\frac{s}{(s+4)^2}$.
- 8 Find a_0 in the Fourier expansion of $f(x) = x + x^2$ in $(-\pi, \pi)$.
- 9 What is cosine half-range Fourier series of a function $f(x)$.
- 10 What is Clairant's equation ? Solve : $y = px + \sin^{-1}(p)$.

(10 × 2 = 20 marks)

Turn over

II. Answer any *six* of the following :

- 11 Derive the necessary and sufficient conditions for the differential equation $Mdx + Ndy = 0$ be exact.
- 12 Solve : $(D^3 + 2D^2 + 4D + 8)y = x^2$.
- 13 Solve : $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$.
- 14 Solve by Charpit's method : $(p^2 + q^2)x = pz$.
- 15 State and prove convolution theorem.
- 16 Find : (i) $L^{-1} \left[\frac{3s+7}{s^2+2s-3} \right]$; (ii) $L^{-1} \left[\frac{s+12}{s^2+4s} \right]$.
- 17 Obtain the Fourier series of $f(x) = |x|$ in $(-1, 1)$.
- 18 Obtain the half-range sine series for $f(x) = x^2$.

(6 × 5 = 30 marks)

III. Answer any *three* of the following :

- 19 (a) With usual notations prove that :

$$\frac{1}{f(D^2)} (\sin(ax)) = \frac{1}{f(-a^2)} (\sin(ax)). \text{ Provided } f(-a^2) \neq 0.$$

(b) Solve : $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 2e^{-3x} + e^x$.

- 20 (a) Explain Charpit's Method for solving partial differential equation $f(x, y, z, p, q) = 0$.

(b) Solve : $\frac{\partial^2 z}{\partial x^2} - 3\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = e^{2x+3y}$.

- 21 (a) Prove that the Laplace transform of the periodic function $f(t)$ with period T is given by :

$$L[f(t)] = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

- (b) Solve by Laplace transform $9y'' - 6y' + y = 0$ given $y(0) = 3$ and $y'(0) = 1$.

- 22 (a) Obtain the Fourier Series of $f(x) = 2x - 1$ in $-1 < x < 1$.

- (b) Find the Fourier finite sine transform for

$$f(x) = 2x - x^2 \text{ in } (0, \pi).$$

(3 × 10 = 30 marks)